

TRIVANDRUM INTL, India

WMO: 433720

Lat: 8.482N

Lon: 76.920E

Elev: 5

StdP: 101.27

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.9	23.8	17.1	12.2	26.4	18.8	13.6	26.6	8.0	28.4	7.4	28.9	1.0	120	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	5.8	33.9	27.6	33.1	27.2	32.9	27.0	28.3	32.2	27.9	31.8	27.6	31.4	4.5	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	22.9	30.9	27.0	22.7	30.8	26.2	21.7	30.3	90.8	32.4	89.0	32.1	87.8	31.9	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	6.0	5.4	DB	21.3	35.0	0.7	1.1	20.7	35.8	20.3	36.4	19.9	37.0	19.4	37.8	(4)
(5)			WB	18.1	29.3	1.7	0.6	16.9	29.8	15.9	30.1	15.0	30.5	13.8	30.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	27.5	28.3	29.4	30.1	29.2	27.9	27.5	27.6	27.7	28.1	28.1	28.0	(6)	
	DBStd	1.22	0.89	0.98	0.93	1.01	1.25	1.04	0.93	0.92	0.77	0.71	0.75	0.70	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6669	542	512	601	603	594	538	544	545	532	561	542	557	(10)	
	CDD18.3	3628	283	278	343	353	336	288	286	287	282	302	292	299	(11)	
	CDH23.3	42219	3249	3431	4499	4817	4271	3117	2904	2889	3029	3360	3260	3393	(12)	
(13)	CDH26.7	16189	1243	1444	2121	2454	1916	981	817	797	907	1128	1140	1243	(13)	
(14)	Wind	WSAvg	2.4	1.6	1.8	2.1	2.6	2.9	3.4	3.4	2.9	2.1	1.5	1.6	(14)	
(15) Precipitation	PrecAvg	1780	19	24	38	119	203	302	203	150	180	276	207	64	(15)	
	PrecMax	2420	162	128	313	309	593	872	421	526	499	656	722	344	(16)	
	PrecMin	1127	0	0	0	5	7	58	47	21	4	17	15	0	(17)	
	PrecStd	319	30	34	48	69	127	167	90	100	136	146	130	69	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	33.2	34.2	34.9	33.8	32.2	32.0	32.1	32.0	32.2	32.2	32.2	(19)	
		MCWB	25.3	26.1	26.9	28.2	27.8	26.9	26.8	26.7	26.8	26.2	26.3	25.5	(20)	
	2%	DB	32.0	32.8	33.2	34.1	33.1	31.9	31.1	31.1	31.2	31.8	31.9	31.9	(21)	
		MCWB	25.5	26.0	26.8	27.9	27.5	26.7	26.7	26.6	26.6	26.2	26.3	25.7	(22)	
	5%	DB	31.2	32.1	33.0	33.2	32.8	31.1	30.8	30.8	30.9	31.1	31.2	31.2	(23)	
		MCWB	25.2	25.5	26.7	27.6	27.3	26.8	26.6	26.5	26.5	26.4	26.3	25.9	(24)	
	10%	DB	31.0	31.8	32.2	33.0	32.1	30.2	30.0	30.0	30.1	30.8	31.0	31.0	(25)	
		MCWB	25.1	25.4	26.3	27.5	26.9	26.4	26.2	26.1	26.2	26.3	26.3	25.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.8	27.4	28.2	29.0	28.7	27.9	27.7	27.3	28.1	27.7	27.7	27.5	(27)	
		MCDB	30.9	31.9	32.7	33.6	32.2	30.2	30.6	30.7	29.7	30.1	30.3	30.5	(28)	
	2%	WB	26.5	26.7	27.6	28.4	28.1	27.3	27.0	26.7	27.1	27.1	27.0	26.9	(29)	
		MCDB	30.6	31.2	32.1	33.0	31.9	30.1	30.4	30.1	30.0	30.3	30.6	30.4	(30)	
	5%	WB	26.2	26.5	27.2	28.0	27.7	26.9	26.7	26.5	26.7	26.7	26.7	26.6	(31)	
		MCDB	30.2	31.0	31.6	32.6	31.4	29.9	29.9	29.8	29.7	30.0	30.2	30.1	(32)	
	10%	WB	25.8	26.1	26.8	27.7	27.3	26.6	26.3	26.2	26.4	26.5	26.5	26.3	(33)	
		MCDB	29.8	30.5	31.2	32.0	31.0	29.5	29.4	29.3	29.4	29.7	29.9	29.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	7.0	6.5	5.8	5.5	4.9	5.0	5.1	5.0	5.4	5.8	6.2	(35)	
		MCDBR	7.5	7.3	7.0	6.1	5.9	5.6	5.8	5.9	5.7	6.0	6.3	6.6	(36)	
	5% WB	MCWBR	3.9	3.8	3.2	3.0	2.6	2.8	3.3	3.2	2.8	2.9	3.0	3.3	(37)	
		MCWBR	6.8	6.6	6.4	6.0	5.5	5.1	5.4	5.5	5.3	5.6	5.9	6.1	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.476	0.500	0.569	0.577	0.577	0.531	0.550	0.526	0.483	0.517	0.527	0.497	(40)	
		taud	2.202	2.129	1.960	1.962	1.970	2.107	2.035	2.117	2.256	2.147	2.097	2.164	(41)	
	Ebn at Noon	Ebn at Noon	838	829	776	760	743	768	757	789	834	805	785	809	(42)	
		Edh at Noon	146	161	193	190	184	158	171	161	141	156	161	149	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	5.47	6.15	6.33	5.97	5.44	5.00	5.12	5.41	5.63	5.12	4.44	4.79	(44)	
	RadStd	RadStd	0.25	0.24	0.37	0.29	0.44	0.41	0.34	0.38	0.39	0.34	0.28	0.30	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (2 neighbors)	+0.28	N/A	N/A	+0.55	+0.60	+0.55	N/A	N/A	N/A	(47)

Nomenclature: See separate page